

Redpwn 2020 Crypto - base646464

deuterium — 2020-06-27

Canonical HTML: <https://deut-erium.github.io/WriteUps/2020/redpwn/crypto/base646464/2020-06-27-redpwn-2020-base646464.html>

Abstract

CTF writeup: redpwnCTF 2020's base646464 applies Base64 25 times. A Python decode loop respectfully declines to call that encryption.

base646464

Description

Encoding something multiple times makes it exponentially more secure!

Files

- [cipher.txt](#)
- [generate.js](#)

Lets take a look at generate.js, which reads

```
const btoa = str => Buffer.from(str).toString('base64');

const fs = require("fs");
const flag = fs.readFileSync("flag.txt", "utf8").trim();

let ret = flag;
for(let i = 0; i < 25; i++) ret = btoa(ret);

fs.writeFileSync("cipher.txt", ret);
```

The function btoa is basically string to its base64 encoding.

The flag is encoded repetitively 25 times in a for loop.

We just need to base64 decode it 25 times and we will get the flag.

```
from base64 import b64decode

with open('cipher.txt','r') as cipher_file:
    data = cipher_file.read()

for i in range(25):
    data = b64decode(data)

print(data)
```

A cool sarcasm on bad crypto challenges in CTFs :)

flag-100ks`11ke`a`10t`of`64s`

| Interactive controls are omitted from this PDF. See the canonical HTML version.